



Faculty of Engineering and the Built Environment
Department of Electrical Engineering

Course Handout: EEE3094S 2026

Course Name	Control Systems Engineering
SAQA Credits	16
Pre-requisites	EEE2047S, MAM2084F
Co-requisites	None

Course convenor	Dr. Stacey Shield
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Office location	Menzies, Room 6.15
Consultation hours	TBA
Course lecturer(s)	Dr. Stacey Shield, Gavin Foster
Teaching assistants	Mpilo Nxasana

Lecture venue	Jordan 3A, tutorials: Jordan 3A
Lecture days and time	Monday, Thursday 10h00-12h00. Tutorials: Wednesday 14h00-16h00
Laboratory venue	Control Laboratory (temporary location, CRG lab)
Laboratory days and times	Book individually

Course objectives

This course teaches the fundamentals of feedback control theory and controller design for linear and time invariant (LTI) systems. We start by recapping the modelling of LTI systems, and then develop concepts of stability and performance for closed-loop systems using the root locus and frequency domain methods. This leads into proportional control, and the design of first-order compensators. Following this course, a student should be able to model an LTI system and design a simple controller to satisfy stability, robustness and performance specifications.

Learning outcomes

Students successfully completing this course will have the following:	Exit level	LO 1	LO 2	LO 3	LO 4	LO 5	LO 6	LO 7	LO 8	LO 9	LO 10	LO 11
A. Knowledge (Information plus understanding)												
LTI systems, system modelling, basic system identification, characteristics of open- and closed-loop systems, theory of continuous-time LTI systems, feedback controller design methods.	N	7	7	7	7	7			7			
B. Skills (Application of knowledge)												
Create and simplify block diagram models of systems. Model simple electro-mechanical systems using differential equations. Derive transfer function models from block diagrams and differential equations. Identify system parameters for first- and second-order LTI systems from time and frequency responses. Analyse closed-loop systems from the transfer function using the root locus, Bode plot, Nyquist plot, Nichols and inverse Nichols charts. Design proportional controllers and first-order compensators (integral, PI, lead and lag.)	N	7	7	7	7	7			7			
C. Values and attitudes												
Appreciation of the translation of fundamental mathematical theory to application through an iterative design process.	N	7	7	7	7	7			7			

Detailed course content

Module 1: Revision of convolution, Laplace transform. Modelling of systems using block diagrams. Derivation of transfer function models from block diagrams and differential equations. Models of simple electromechanical systems. Characteristics of first- and second-order LTI systems.

Module 2: Analysis of closed-loop systems and proportional controller design using the root locus, Nyquist plot, Nichols chart and inverse Nichols chart. Closed-loop stability and performance. Proportional controller design.

Module 3: Design and combination of first-order compensators (integrator, PI, lead, lag).

Knowledge areas

Maths Sciences	Natural Sciences	Engineering Sciences	Design & Synthesis	Complementary Studies
		50	50	

Learning environment

Lectures, tutorials, laboratories

Suggested time allocation

Learning Activity	No./ week	Time in hours	Contact time Multiplier	Total no of hours
Number of lectures per week	4	0.75	2	6x12=72
Number of tutorials per week	1	2	2	2x12=24
Total practical/lab periods	0.25	2	2	1x12=12
Total other contact periods				
Total assignment non-contact hours (CS)				30
Assessment hours (Tests, Exam)				22
Number of weeks the course lasts	12			
Total hours				160

General assessment strategy

Assessment Task	%	The following DP rules apply:
Assignments		Students must: 1) submit labs 1 and 2. Lab 3 is an optional extra credit assignment that will be added to the lab score, up to a total of 100%. Labs have a late penalty of 20% per day.
Tutorials		
Labs	$(2 \times 0.375 + 0.375, \text{ capped } 0.75) \times 20/40$	
Project		Repeating students who have passed the labs before may apply for an exemption, in which case they will carry these marks over from the previous attempt. 2) Attend 90% of lectures and tutorials during designated teaching weeks of the course. Attendance will be monitored using the 'Kam Cam' selfie register. Note: students do not have to excuse themselves from class tests. The written assessment portion of the class mark will be calculated by taking the best two marks out of the following 4 marks: (1) class test 1 (2) class test 2 (3) aggregate of best 4 tutorial tests and (4) the exam.
Tests (i.e. written assessment)	$2 \times 0.125 \times 20/40$	
Exam	80/60 in student's favour	
Total		

Books/Reading Materials/Notes

N. S. Nise, Control Systems Engineering, 5th Ed, John Wiley & Sons Inc., Asia, 2008.

Absence: The continuous assessment marks will be adjusted to allow for absence only on the following grounds:

A medical certificate for absence of 3 or more consecutive days or any day with a class test or exam

Death of an immediate family member (parent or sibling)

Pre-arranged absence to represent a University, provincial or national team.

Academic dishonesty: Plagiarism is a very serious offence and usually leads to disciplinary action that could include expulsion from the university. Therefore, recognise the work of others in any submission. Details of referencing methods are widely available on the Web. A non-plagiarism declaration must be submitted with all work submitted for marking.